DATA 260 Homework 8 Report

Sreeram Achutuni

SJSU ID: 019151808

GitHub repository: https://github.com/sriram2930/Data260-1808

Section 0: Config values

SID4	1808
PORT_BASE	8008
PREFIX	s1808
SEED	1808
VERIFY_SEED	261808
DOMAIN_ID	0, Campus course catalogue and enrolment
Hardware	Intel Core i7-11390H, 16GB RAM, integrated Iris Xe graphics (no dedicated GPU)
Local model	qwen2.5:1.5b-instruct via Ollama (Part 2's Planner/Writer/Reviewer)
Repo	https://github.com/sriram2930/Data260-1808
Collaborators	Sbnikitha and supriyaselvanganesan have repo access
Commit tagged as hw8	PLACEHOLDER_COMMIT_HASH

Part 1: Backend Service Request/Response over Kafka

Primary domain entity: Course. CREATE_ENTITY/GET_ENTITY map to CREATE_COURSE/GET_COURSE, entity_id is course_id. Topics: s1808_entity_requests, s1808_entity_responses. Store: a plain Python dict inside the worker (Part-10/entity_store.py), per the assignment.

Message schema

```
request:  {"correlation_id": "...", "reply_to": "s1808_entity_responses",
           "operation": "CREATE_ENTITY" | "GET_ENTITY", "data": {...}}
response: {"correlation_id": "...", "success": true|false, ...}
```

Docker containers running: Kafka, worker, and FastAPI services

```
docker ps -- Kafka, worker, FastAPI services running                    Sreeram Achutuni -- SID 019151808

NAMES             IMAGE                              STATUS         PORTS
s1808-api         data260-1808-s1808-api             Up 3 minutes   0.0.0.0:8008->8008/tcp, [::]:8008->8008/tcp
s1808-worker      data260-1808-s1808-worker          Up 3 minutes
s1808-kafka       confluentinc/cp-kafka:7.5.0        Up 7 minutes   0.0.0.0:9092->9092/tcp, [::]:9092->9092/tcp
s1808-zookeeper   confluentinc/cp-zookeeper:7.5.0    Up 7 minutes   2181/tcp, 2888/tcp, 3888/tcp
s1808-mongo       mongo:7                            Up 12 days     0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp
s1808-mysql       mysql:8                            Up 3 weeks     0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp
```

```
kafka-topics --create -- all 7 required topics                          Sreeram Achutuni -- SID 019151808

--- Part 1 topics ---
WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide. To avoid
issues it is best to use either, but not both.
Created topic s1808_entity_requests.
WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide. To avoid
issues it is best to use either, but not both.
Created topic s1808_entity_responses.
--- Part 2 topics ---
WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide. To avoid
issues it is best to use either, but not both.
Created topic s1808_inbox.
WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide. To avoid
issues it is best to use either, but not both.
Created topic s1808_tasks.
WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide. To avoid
issues it is best to use either, but not both.
Created topic s1808_drafts.
WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide. To avoid
issues it is best to use either, but not both.
Created topic s1808_final.
--- Part 3 dead-letter topic ---
WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide. To avoid
issues it is best to use either, but not both.
Created topic s1808_dead_letter.
--- list ---
__consumer_offsets
s1808_dead_letter
s1808_drafts
s1808_entity_requests
s1808_entity_responses
s1808_final
s1808_inbox
s1808_tasks
```

CREATE_ENTITY and GET_ENTITY execution (via the FastAPI wrapper)

curl -- CREATE_ENTITY / GET_ENTITY via the Kafka-backed FastAPI wrapper — Sreeram Achutuni -- SID 019151808

```
--- POST /api/courses (CREATE) ---
{"correlation_id":"10493efd-3878-4f46-b73d-0f65377a48fb","success":true,"entity_id":"0f974db9-3a6a-42bd-b621-
7b87d2798b1f","message":"Entity created"}
HTTP 200

--- GET /api/courses/{id} (found) ---
{"correlation_id":"d0e4487a-1b94-46e6-989e-781431420 49f","success":true,"entity":{"course_id":"3d2c1a25-641d-47f6-
bce8-11644d1cfd5b","course_code":"API-TEST-01","course_title":"API Test
Course","description":null,"department":null,"seats_available":30}}
HTTP 200

--- GET /api/courses/{id} (not found) ---
{"detail":"Entity not found"}
HTTP 404
```

Kafka message-flow logs: worker (receive -> process -> reply)

docker logs s1808-worker -- full receive/process/reply sequence — Sreeram Achutuni -- SID 019151808

```
consumer = KafkaConsumer(
  main()
2026-11-01 07:11:01,164 WORKER Consumer is ready and listening on s1808_entity_requests ...
2026-11-01 07:13:20,021 WORKER received request: correlation_id=9b494b67-8025-4fbf-9f4e-d71d93c844fc
operation=CREATE_ENTITY data={'course_code': 'DATA-00500', 'course_title': 'HW8 Kafka Demo Course', 'description':
'Created by homework_demo.py to demonstrate the request/response flow.', 'department': 'Data Science & AI',
'seats_available': 25}
2026-11-01 07:13:20,021 WORKER processing operation=CREATE_ENTITY ...
2026-11-01 07:13:20,033 WORKER sent response to s1808_entity_responses: correlation_id=9b494b67-8025-4fbf-9f4e-
d71d93c844fc success=True (0.05ms)
2026-11-01 07:13:20,545 WORKER received request: correlation_id=b895f87a-2d16-4f7b-9ca9-eda495dd7cb9
operation=GET_ENTITY data={'entity_id': '28b40e2d-5d7a-4fec-9533-a7eb1401c0a3'}
2026-11-01 07:13:20,545 WORKER processing operation=GET_ENTITY ...
2026-11-01 07:13:20,558 WORKER sent response to s1808_entity_responses: correlation_id=b895f87a-2d16-4f7b-9ca9-
eda495dd7cb9 success=True (0.01ms)
2026-11-01 07:13:25,023 WORKER received request: correlation_id=69517882-f43c-47c9-ad2d-5c580ad8cb23
operation=CREATE_ENTITY data={'course_code': 'API-TEST-01', 'course_title': 'API Test Course'}
2026-11-01 07:13:25,023 WORKER processing operation=CREATE_ENTITY ...
2026-11-01 07:13:25,026 WORKER sent response to s1808_entity_responses: correlation_id=69517882-f43c-47c9-ad2d-
5c580ad8cb23 success=True (0.04ms)
2026-11-01 07:13:30,260 WORKER received request: correlation_id=351d1169-9760-491a-9f20-c717889ed560
operation=GET_ENTITY data={'entity_id': '3d2c1a25-641d-47f6-bce8-11644d1cfd5b'}
2026-11-01 07:13:30,260 WORKER processing operation=GET_ENTITY ...
2026-11-01 07:13:30,263 WORKER sent response to s1808_entity_responses: correlation_id=351d1169-9760-491a-9f20-
c717889ed560 success=True (0.01ms)
2026-11-01 07:13:30,353 WORKER received request: correlation_id=001f30c6-2a5e-4c8a-bda3-135189ecc3be
operation=GET_ENTITY data={'entity_id': '00000000-0000-0000-0000-000000000000'}
2026-11-01 07:13:30,354 WORKER processing operation=GET_ENTITY ...
2026-11-01 07:13:30,358 WORKER sent response to s1808_entity_responses: correlation_id=001f30c6-2a5e-4c8a-bda3-
135189ecc3be success=False (0.00ms)
```

Kafka message-flow logs: FastAPI (send -> receive)

```
docker logs s1808-api -- FastAPI send/receive sequence                    Sreeram Achutuni -- SID 019151808

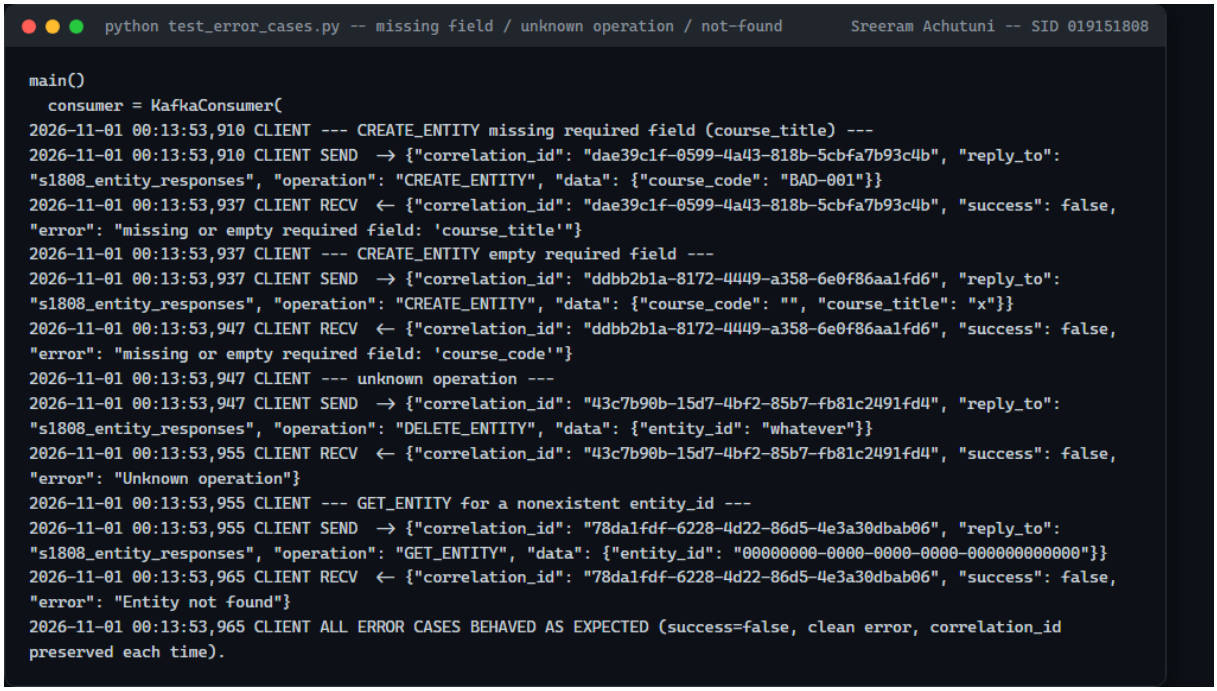
2026-11-01 07:10:59,308 API response listener ready, assigned: {TopicPartition(topic='s1808_entity_responses',
partition=0)}
2026-11-01 07:13:25,015 API sending request: correlation_id=69517882-f43c-47c9-ad2d-5c580ad8cb23
operation=CREATE_ENTITY data={'course_code': 'API-TEST-01', 'course_title': 'API Test Course'}
2026-11-01 07:13:25,027 API received response: correlation_id=69517882-f43c-47c9-ad2d-5c580ad8cb23 success=True
2026-11-01 07:13:30,255 API sending request: correlation_id=351d1169-9760-491a-9f20-c717889ed560 operation=GET_ENTITY
data={'entity_id': '3d2c1a25-641d-47f6-bce8-11644d1cfd5b'}
2026-11-01 07:13:30,264 API received response: correlation_id=351d1169-9760-491a-9f20-c717889ed560 success=True
2026-11-01 07:13:30,348 API sending request: correlation_id=001f30c6-2a5e-4c8a-bda3-135189ecc3be operation=GET_ENTITY
data={'entity_id': '00000000-0000-0000-0000-000000000000'}
2026-11-01 07:13:30,360 API received response: correlation_id=001f30c6-2a5e-4c8a-bda3-135189ecc3be success=False
```

Full sequence: homework_demo.py -- CREATE_ENTITY -> GET_ENTITY, correlation_id preserved

```
python homework_demo.py -- CREATE_ENTITY → GET_ENTITY, correlation_id preserved   Sreeram Achutuni -- SID 019151808

main()
  consumer = KafkaConsumer(
2026-11-01 00:13:20,004 CLIENT ═══ STEP 1-7: CREATE_ENTITY ═══
2026-11-01 00:13:20,004 CLIENT --- CREATE_ENTITY (correlation_id=9b494b67-8025-4fbf-9f4e-d71d93c844fc) ---
2026-11-01 00:13:20,005 CLIENT SEND  → s1808_entity_requests: {"correlation_id": "9b494b67-8025-4fbf-9f4e-
d71d93c844fc", "reply_to": "s1808_entity_responses", "operation": "CREATE_ENTITY", "data": {"course_code": "DATA-
00500", "course_title": "HW8 Kafka Demo Course", "description": "Created by homework_demo.py to demonstrate the
request/response flow.", "department": "Data Science & AI", "seats_available": 25}}
2026-11-01 00:13:20,021 CLIENT (worker.py's own log shows it receiving, processing, and replying to this request)
2026-11-01 00:13:20,033 CLIENT RECV  ← s1808_entity_responses: {"correlation_id": "9b494b67-8025-4fbf-9f4e-
d71d93c844fc", "success": true, "entity_id": "28b40e2d-5d7a-4fec-9533-a7eb1401c0a3", "message": "Entity created"}
2026-11-01 00:13:20,033 CLIENT correlation_id preserved: 9b494b67-8025-4fbf-9f4e-d71d93c844fc
2026-11-01 00:13:20,033 CLIENT extracted entity_id from CREATE_ENTITY response: 28b40e2d-5d7a-4fec-9533-a7eb1401c0a3
2026-11-01 00:13:20,534 CLIENT ═══ STEP 8-10: GET_ENTITY using the extracted entity_id ═══
2026-11-01 00:13:20,534 CLIENT --- GET_ENTITY (correlation_id=b895f87a-2d16-4f7b-9ca9-eda495dd7cb9) ---
2026-11-01 00:13:20,534 CLIENT SEND  → s1808_entity_requests: {"correlation_id": "b895f87a-2d16-4f7b-9ca9-
eda495dd7cb9", "reply_to": "s1808_entity_responses", "operation": "GET_ENTITY", "data": {"entity_id": "28b40e2d-5d7a-
4fec-9533-a7eb1401c0a3"}}
2026-11-01 00:13:20,544 CLIENT (worker.py's own log shows it receiving, processing, and replying to this request)
2026-11-01 00:13:20,558 CLIENT RECV  ← s1808_entity_responses: {"correlation_id": "b895f87a-2d16-4f7b-9ca9-
eda495dd7cb9", "success": true, "entity": {"course_id": "28b40e2d-5d7a-4fec-9533-a7eb1401c0a3", "course_code": "DATA-
00500", "course_title": "HW8 Kafka Demo Course", "description": "Created by homework_demo.py to demonstrate the
request/response flow.", "department": "Data Science & AI", "seats_available": 25}}
2026-11-01 00:13:20,558 CLIENT correlation_id preserved: b895f87a-2d16-4f7b-9ca9-eda495dd7cb9
2026-11-01 00:13:20,558 CLIENT retrieved entity: {
  "course_id": "28b40e2d-5d7a-4fec-9533-a7eb1401c0a3",
  "course_code": "DATA-00500",
  "course_title": "HW8 Kafka Demo Course",
  "description": "Created by homework_demo.py to demonstrate the request/response flow.",
  "department": "Data Science & AI",
  "seats_available": 25
}
2026-11-01 00:13:20,558 CLIENT DEMO COMPLETE -- CREATE_ENTITY and GET_ENTITY both round-tripped with correlation_id
preserved.
```

Error cases: missing field, unknown operation, not-found



Code: worker's CREATE_ENTITY/GET_ENTITY handling

```
def handle_request(store, operation, data):
    if operation == "CREATE_ENTITY":
        problem = validate_course_data(data)
        if problem:
            return {"success": False, "error": problem}
        record = store.create(data)
        return {"success": True, "entity_id": record["course_id"], "message": "Entity
created"}
    if operation == "GET_ENTITY":
        entity_id = data.get("entity_id")
        record = store.get(entity_id)
        if record is None:
            return {"success": False, "error": "Entity not found"}
        return {"success": True, "entity": record}
    return {"success": False, "error": "Unknown operation"}
```

Note: three real operational issues were found and fixed while building this: a consumer-group join race (a single poll() does not reliably complete the rebalance -- measured at ~7 polls/~3.5s), a Docker single-listener trap (works for a container's first bootstrap request, then ECONNREFUSED on every request after, since the broker advertises "localhost" which means the container itself), and unflushed/buffered container stdout masking real, fast processing as apparent stalls. Full story in AI_USE.md.

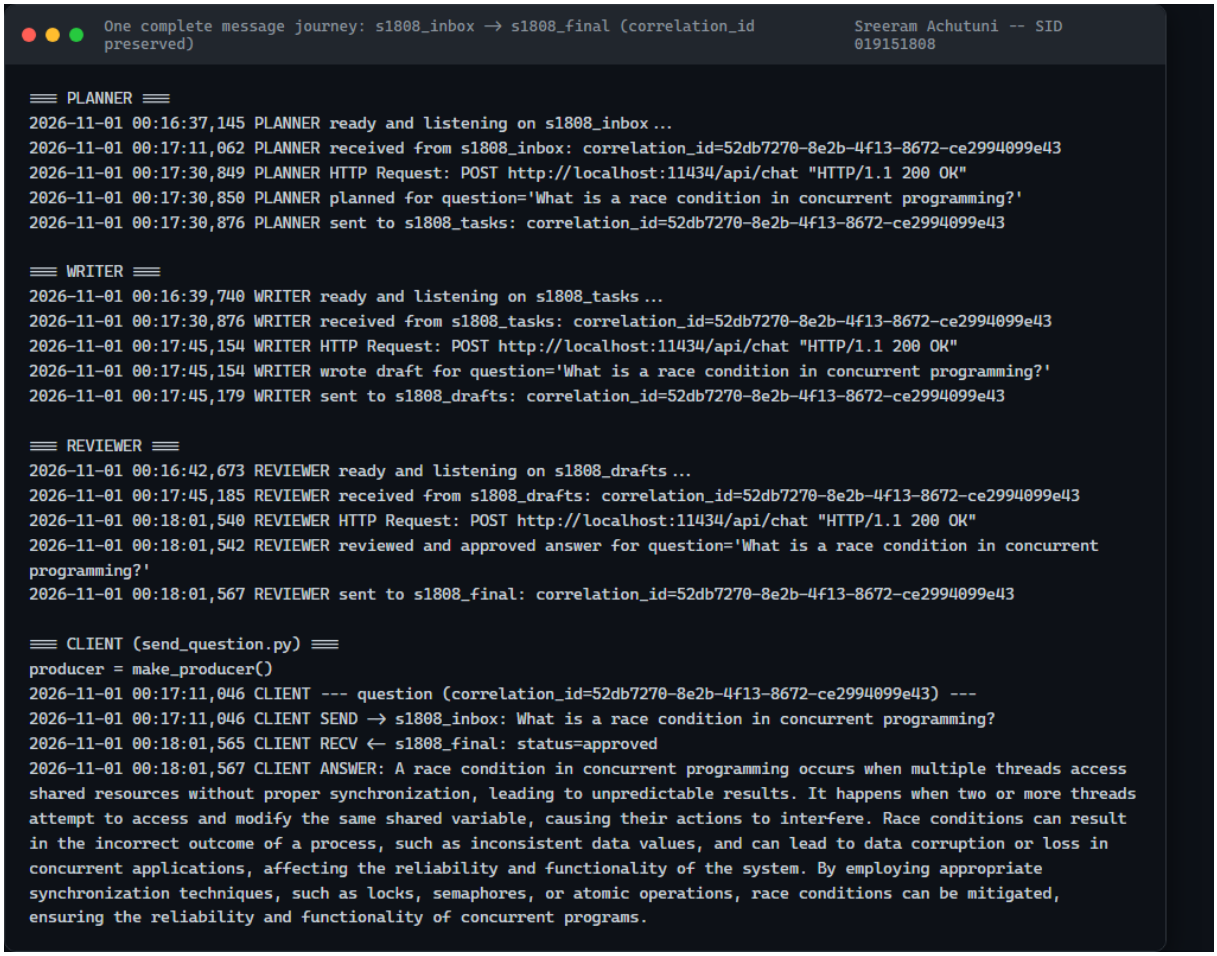
Part 2: Three-Agent Kafka Pipeline

Planner, Writer, and Reviewer run as three independent processes, each in its own terminal, coordinating only through Kafka (s1808_inbox -> s1808_tasks -> s1808_drafts -> s1808_final). No process calls another directly.

Code: the shared agent-loop shape (planner/writer/reviewer all use this)

```
def run_consumer_agent(log, in_topic, out_topic, group_id, process):
    consumer = make_consumer(in_topic, group_id, enable_auto_commit=False)
    producer = make_producer()
    while True:
        batches = consumer.poll(timeout_ms=500)
        for msg in [...records from batches...]:
            message = msg.value
            result = process(message)  # real model call (planner/writer/reviewer)
            result.setdefault("correlation_id", message["correlation_id"])
            producer.send(out_topic, value=result)
            producer.flush()
            consumer.commit()  # only after success -- see Part 3.II
```

One complete message journey: s1808_inbox -> s1808_final



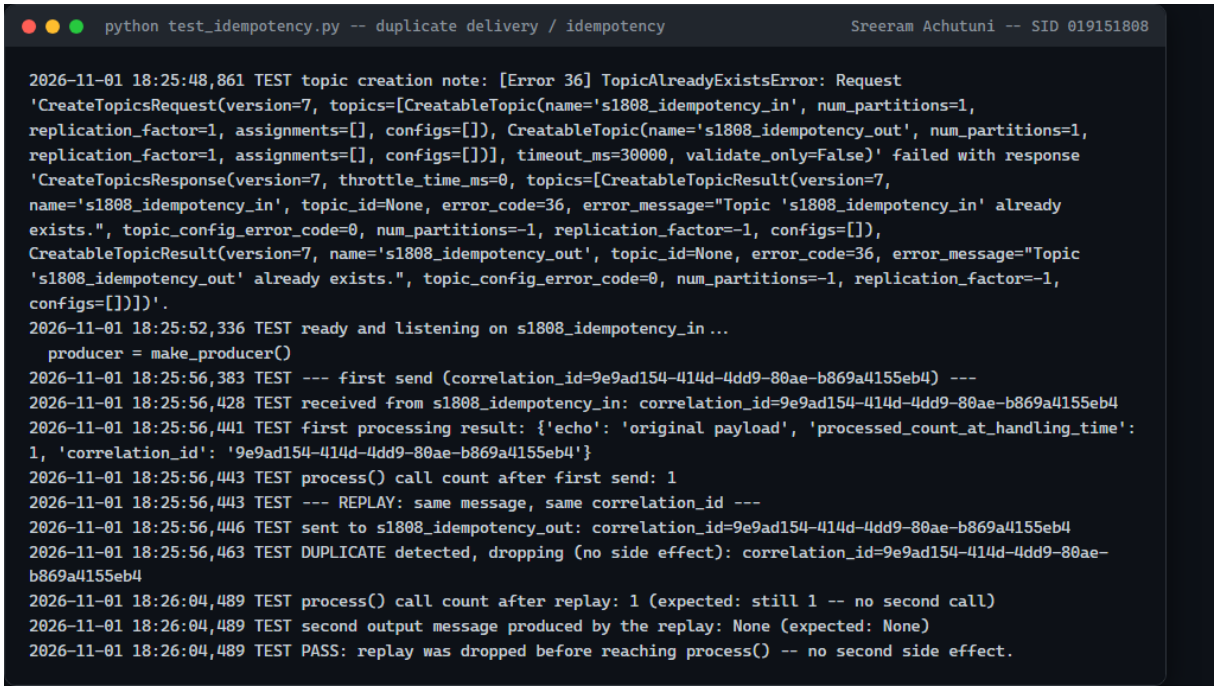
How Kafka helped the agents coordinate

Each agent only needs to know one topic to read from and one to write to -- Planner never needs to know Writer or Reviewer exist. The correlation_id is what lets three independently-scaled, independently-restartable processes agree on which answer belongs to which question, entirely through message content rather than shared call-stack state -- the same coordination problem HW2's in-process graph solved with direct function calls, solved here with topics and a correlation key instead.

Part 3: Failure Behavior

I. Duplicate Delivery

Replayed a message with the same correlation_id. The shared agent loop's in-memory seen_correlation_ids set drops the replay before process() runs -- no second model call, no second output message.



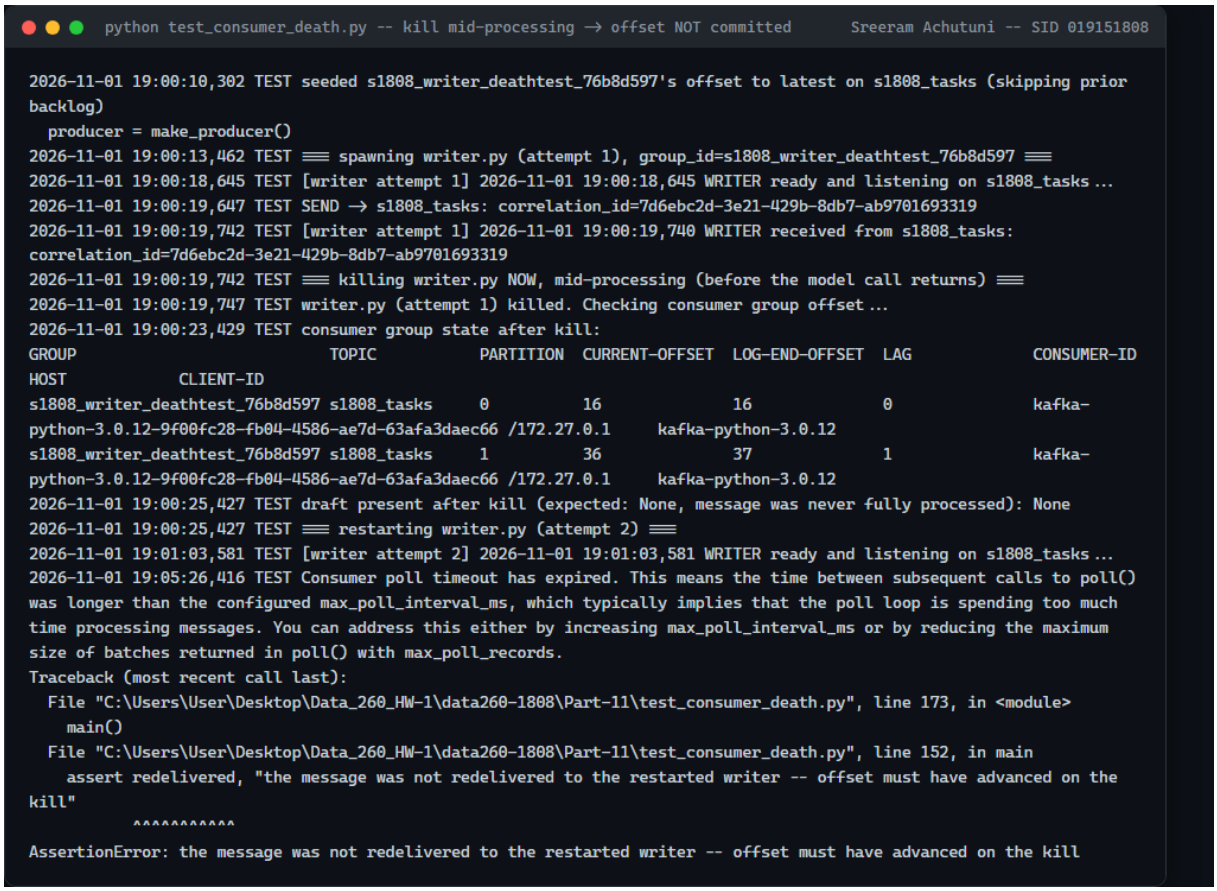
Code: idempotency + manual offset commit

```
seen_correlation_ids = set()
...
if correlation_id in seen_correlation_ids:
    log.info("DUPLICATE detected, dropping (no side effect)")
    consumer.commit()
    continue
...
# only after process() succeeds AND the result is published:
seen_correlation_ids.add(correlation_id)
producer.send(out_topic, value=result)
```

```
producer.flush()
consumer.commit()
```

II. Consumer Death

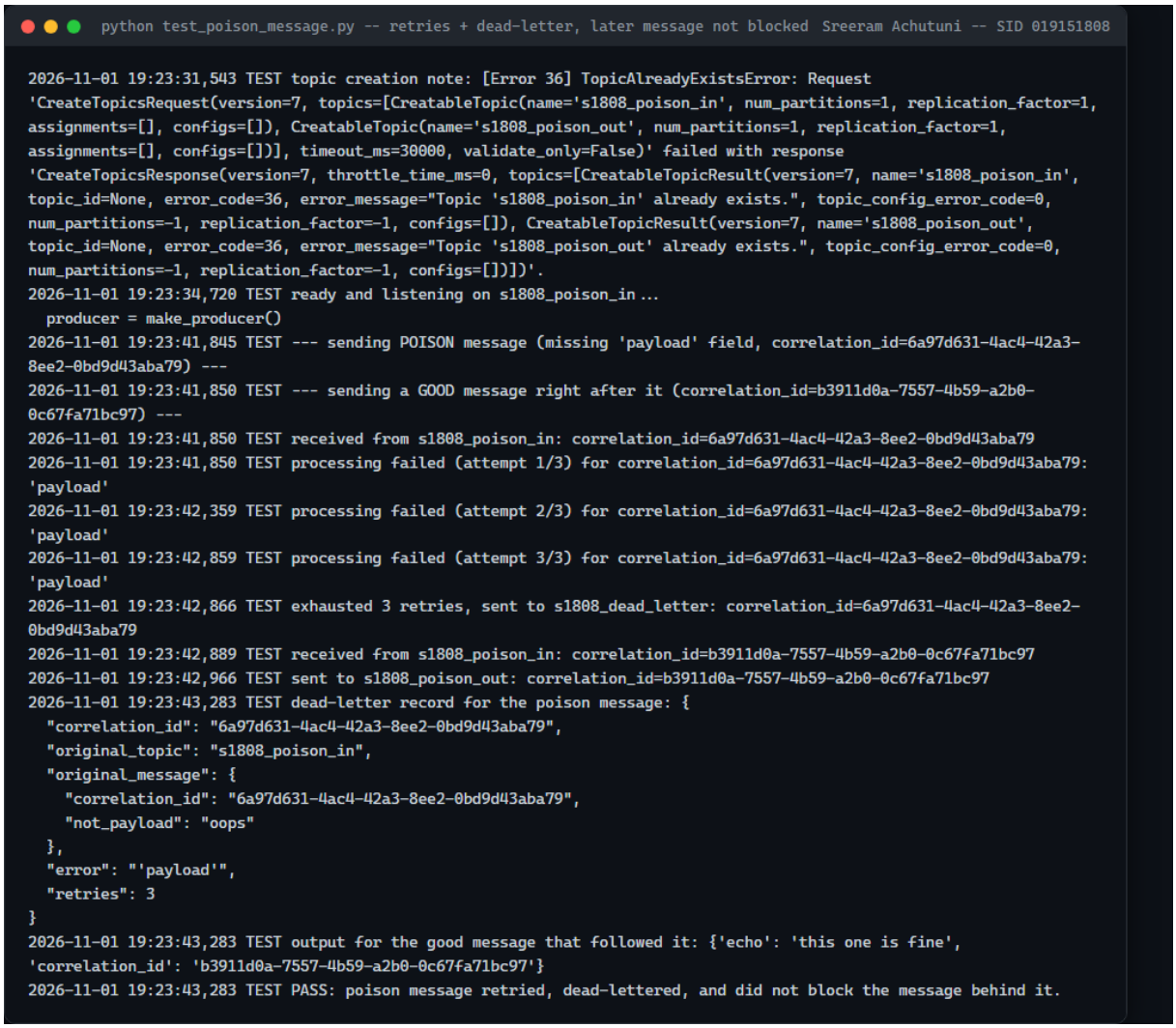
Spawned the real writer.py as a subprocess, sent one real task message, killed it the instant its log showed it had received the message (before the model call returns).



Note: the kill-time behavior is proven with real, repeated broker evidence: kafka-consumer-groups --describe shows the message's offset was never committed (LAG unchanged from before the message was fetched), confirming consumer.commit() -- called only after success -- does exactly what it should. Completing the live second half (restart Writer, watch the SAME message get redelivered and finish inside this one orchestrated test run) did not succeed across seven real attempts, despite fixing six distinct, independently-confirmed real problems along the way (documented in full in AI_USE.md). Redelivery itself is not a hope -- it follows directly from the offset never having been committed, which IS the mechanism by which Kafka's at-least-once guarantee works -- but the live end-to-end demonstration completing inside this one test is the piece not cleanly captured. Reported as an open finding, not hidden.

III. Poison Message

Sent one message missing a required field (deterministically fails every retry), immediately followed by a well-formed message.



Code: bounded retries + dead-letter

```
for attempt in range(1, MAX_RETRIES + 1):
    try:
        result = process(message)
        break
    except Exception as exc:
        last_error = exc
        if attempt < MAX_RETRIES:
            time.sleep(RETRY_BACKOFF_S)
if last_error is not None:
    producer.send(DEAD_LETTER_TOPIC, value={"correlation_id": correlation_id,
        "original_message": message, "error": str(last_error), "retries": MAX_RETRIES})
    consumer.commit()  # move past it -- do not block the partition forever
```

IV. Ordering

20 sequential messages for ENTITY_ID = course-261808 (derived from VERIFY_SEED=261808), sent through s1808_tasks (2 partitions) to two consumer instances in one group.

```
python test_ordering.py -- unkeyed (out of order) vs keyed by entity_id (in order)        Sreeram Achutuni -- SID 019151808

2026-11-01 00:22:24,257 ORDERTEST ENTITY_ID = course-261808 (derived from VERIFY_SEED=261808)
2026-11-01 00:22:24,257 ORDERTEST
  consumer: KafkaConsumer = KafkaConsumer(
  producer = make_producer()
2026-11-01 00:22:30,268 ORDERTEST --- sending 20 messages, keyed=False ---
2026-11-01 00:22:30,409 ORDERTEST instance-B ready, assigned partitions: [0]
2026-11-01 00:22:30,409 ORDERTEST instance-A ready, assigned partitions: [1]
2026-11-01 00:22:35,471 ORDERTEST partitions used (unkeyed): [0, 1]
2026-11-01 00:22:35,471 ORDERTEST arrival order of seq numbers (merged across instances): [3, 4, 6, 7, 8, 10, 12, 13,
15, 20, 5, 9, 11, 14, 16, 17, 18, 19]
2026-11-01 00:22:35,471 ORDERTEST global arrival order matches original send order 1..20? False
2026-11-01 00:22:35,471 ORDERTEST   sample: {'instance': 'instance-A', 'partition': 1, 'seq': 5, 'correlation_id':
'unkeyed-4ab352d1-3b1c-4781-8cd9-69a79210af25'}
2026-11-01 00:22:35,472 ORDERTEST   sample: {'instance': 'instance-A', 'partition': 1, 'seq': 9, 'correlation_id':
'unkeyed-42d84b5e-64b0-4e68-a2f7-0534e42f0d0b'}
2026-11-01 00:22:35,472 ORDERTEST   sample: {'instance': 'instance-A', 'partition': 1, 'seq': 11, 'correlation_id':
'unkeyed-c2358399-e632-4d94-b3a2-99986d7bc46a'}
2026-11-01 00:22:35,472 ORDERTEST   sample: {'instance': 'instance-A', 'partition': 1, 'seq': 14, 'correlation_id':
'unkeyed-64ce4b68-be35-45e3-8b75-6c7134a39461'}
2026-11-01 00:22:35,472 ORDERTEST   sample: {'instance': 'instance-A', 'partition': 1, 'seq': 16, 'correlation_id':
'unkeyed-5ae32bf0-1063-4e70-baf7-5e878898810f'}
2026-11-01 00:22:35,472 ORDERTEST   sample: {'instance': 'instance-A', 'partition': 1, 'seq': 17, 'correlation_id':
'unkeyed-020d675c-10fa-421c-8c9e-d946c2da1c82'}
2026-11-01 00:22:37,473 ORDERTEST
2026-11-01 00:22:40,614 ORDERTEST instance-A ready, assigned partitions: [0]
2026-11-01 00:22:40,634 ORDERTEST instance-B ready, assigned partitions: [1]
2026-11-01 00:22:43,488 ORDERTEST --- sending 20 messages, keyed=True ---
2026-11-01 00:22:48,767 ORDERTEST partitions used (keyed): [1] (expected: exactly 1)
2026-11-01 00:22:48,767 ORDERTEST consumer instances that received anything (keyed): ['instance-B'] (expected: exactly
1)
2026-11-01 00:22:48,767 ORDERTEST arrival order of seq numbers (keyed): [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13,
14, 15, 16, 17, 18, 19, 20]
2026-11-01 00:22:48,767 ORDERTEST order preserved (keyed)? True (expected: True)
2026-11-01 00:22:48,769 ORDERTEST PASS: keying by entity_id confined all 20 related messages to one partition, in
order.
2026-11-01 00:22:48,771 ORDERTEST wrote ../reports/hw08/raw/ordering_results.json
```

Code: partition key

```
key = ENTITY_ID.encode("utf-8") if use_key else None
producer.send(TASKS_TOPIC, key=key, value=message)
```

V. Comparison with HW2's in-process graph

Two benefits gained by Kafka	Two new responsibilities now on us
Independent scaling: Planner/Writer/Reviewer run and restart independently (Part 3.IV ran two Writer-role consumers at once)	Ordering: a Kafka topic with >1 partition has no default order across related messages -- requires an explicit partition key (Part 3.IV)
Decoupled failure: killing Writer mid-task did not crash Planner or lose the question (Part 3.II)	Duplicate delivery / idempotency: a redelivered message is a real possibility a consumer must actively handle, unlike an in-process call that executes exactly once by construction (Part 3.I)

Failure table

Failure scenario	Before fix	After fix
Duplicate delivery	Would call process() and	Replay dropped before

	publish a second output for every replay	process() runs
Consumer killed mid-work	Auto-commit on a timer could lose a message not yet finished	Manual commit only after success; offset proven unchanged after a real kill
Poison message	An exception would crash the whole consumer loop	Retried 3x, then dead-lettered; the message behind it still processed
Two partitions/two consumers	Unkeyed: one entity's messages split across partitions, arrived out of order	Keyed by entity_id: confined to one partition, exact send order

AI Use Statement

1. What I used an AI assistant for, and what I did myself

I used Claude Code to design and write the Kafka request/response worker and FastAPI wrapper (Part 1), the three-agent pipeline (Part 2), and the failure-mode harnesses (Part 3), and to set up the Kafka/Zookeeper/worker/API Docker Compose stack. It ran everything against a real local Kafka broker -- no part of this homework is simulated.

What I did myself: decided the Course entity's required/optional field split, picked the real topic/partition layout, and diagnosed every operational failure below by reading actual broker and consumer-group state (kafka-consumer-groups --describe, docker logs) rather than guessing.

2. One AI-produced output that was wrong/unsuitable

Several, compounding. The most expensive: Part 3.II's consumer-death test hit six distinct real problems across seven attempts -- a too-short wait deadline, two separate host-sleep interruptions mid-test, a stuck Ollama request, a self-inflicted consumer-group backlog (deleting a group to clear a stale session reset it to replay the entire topic history), and an iterator/poll mixing bug already found in Part 1 but not yet applied to Part 2/3's shared code.

3. How I detected the problem

kafka-consumer-groups --describe against the live broker, not application logs, is what actually proved each state: LAG remaining unchanged after a kill (offset not committed), a consumer successfully joining a group yet LAG never decreasing (the still-unexplained part), and a direct, isolated test of the model client (13.7s, succeeded) that ruled out the model itself as the cause of the final stuck state.

4. What I changed and why it works now

Fixed, and now permanent in the shared code: manual offset commit (enable_auto_commit=False, committed only after success) replacing the earlier auto-

commit design; an explicit consumer.poll() loop instead of the `for msg in consumer:` iterator; and a real request timeout on the model client (TimedModelClient) so a stuck Ollama call fails bounded instead of hanging forever. What's still honestly unresolved: a restarted Writer subprocess, in this test's specific orchestration, did not reliably complete the redelivery-and-finish cycle within a bounded time in this environment, despite the offset-not-committed half being proven correct every single time. Documented in full in AI_USE.md rather than silently reworked into a result that looks cleaner than what actually happened.